Fully-Bayesian Inference

Marginalising Over All Hyperparameters

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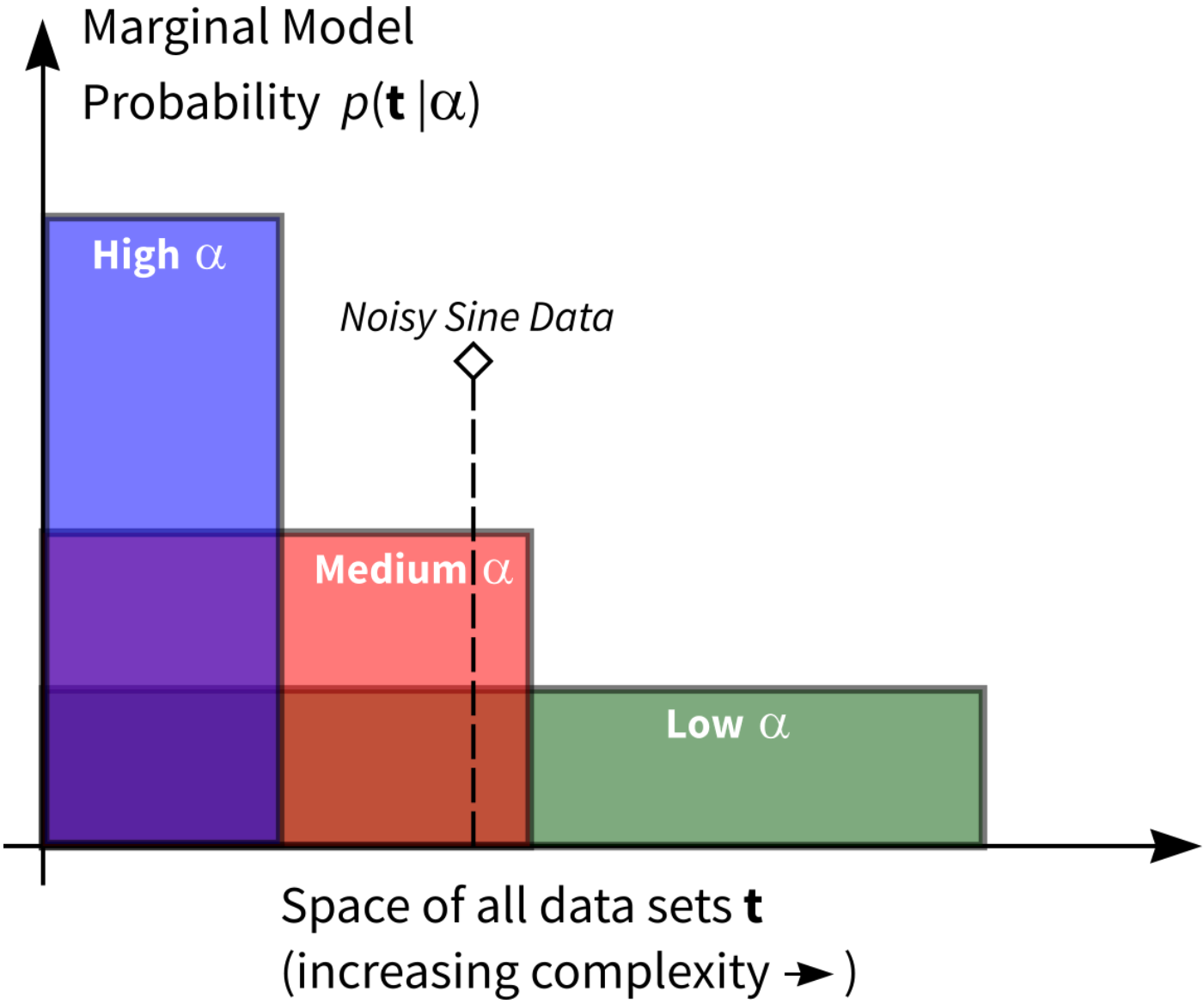
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Slide 1 In this Episode ...

- Recap: the implicit Occam's Razor effect for α (and σ)
- Extending the analysis to more parameters: basis width r
- Mixing marginalisation and Type-II maximum likelihood for different hyperparameters
- Demo:
  - not only estimating r, but also
  - the predictive distribution, and
  - selection of basis function type $\phi(\cdot)$
- This works pretty nicely, but even in this simple model becomes a complicated mess!

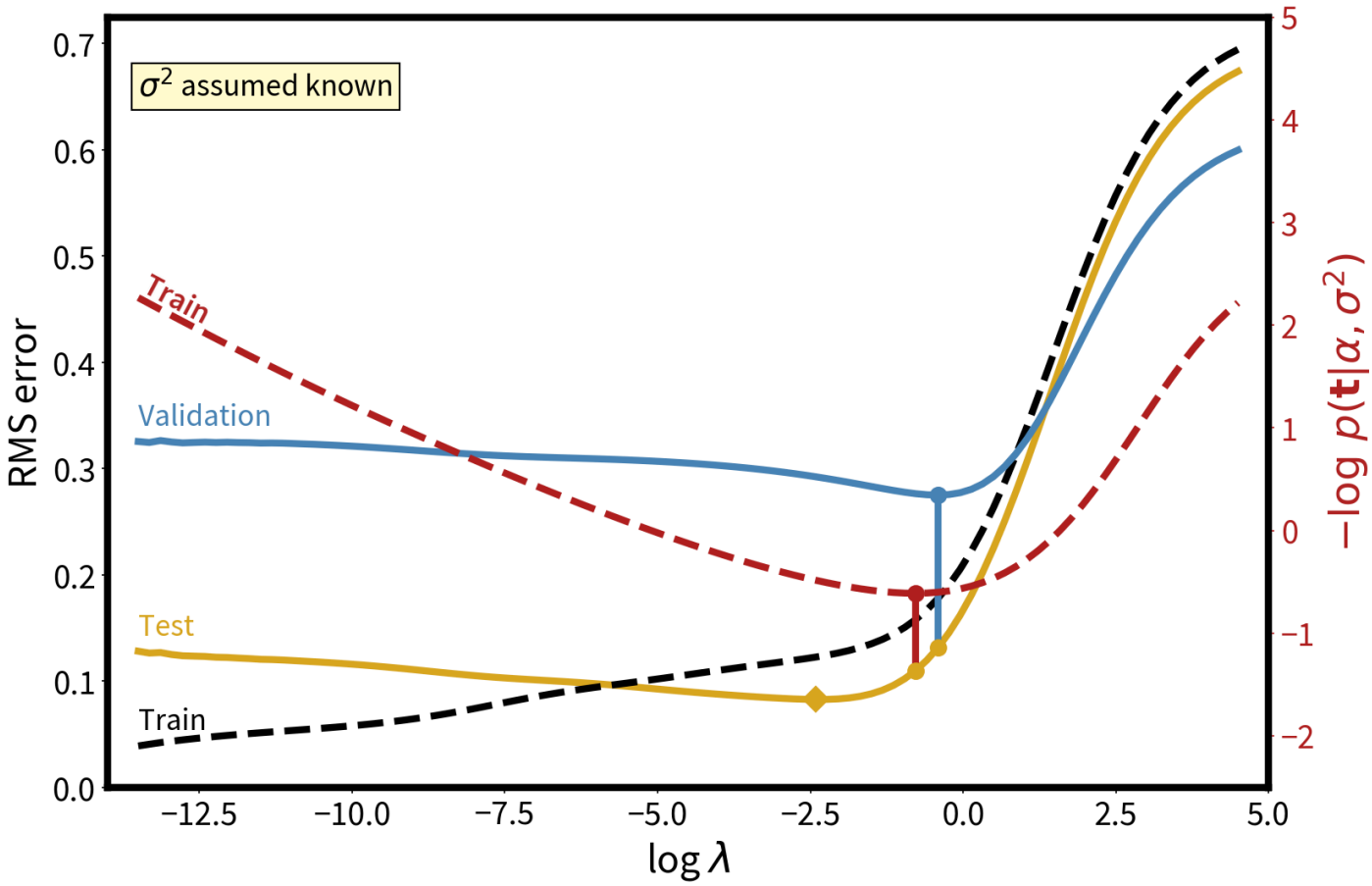
Recap

Slide 2 Implicit Implementation of "Occam's Razor" in $p\left(\mathbf{y}|\alpha\right)$



"Entities should not be multiplied unnecessarily." *William of Occam — 14th Century*

Slide 3 Marginal Likelihood to Estimate Hyper-parameter $\widehat{\alpha}_{MP}$



Slide 4 Strictly Bayesian: the Ideal Approach

In general, for any model, if we wish to predict y_* (at x_*) given some training data $\mathbf{y}$, what we ideally should compute is:

$$p\left(y_*|\mathbf{y}\right)$$

Slide 5 Being Strictly Bayesian for Linear Regression (So Far)

- We placed a prior over the weights $p\left(\mathbf{w}|\alpha\right)$, and defined:
  - $p\left(\alpha\right)$ — a hyper-prior over the hyper-parameter, and

- $p\left(\sigma^2\right)$ — a prior over the noise variance in the observation model

- We would then ideally have computed the full posterior: $p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right)$

- ... and used it to make the desired predictions:

$$p\left(y_* | \mathbf{y}\right) = \int p\left(y_* | \mathbf{w}, \alpha, \sigma^2\right) p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right) d\mathbf{w}\, d\alpha\, d\sigma^2$$

- This prediction incorporates our uncertainty over all unknown parameters by averaging over their likely values having seen the data **y**

Slide 6 **The Analytic Roadblock**

“Strict Bayesian” inference:

- In our model, would ideally be:

$$p\left(y_* | \mathbf{y}\right) = \int p\left(y_* | \mathbf{w}, \alpha, \sigma^2\right) p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right) d\mathbf{w}\, d\alpha\, d\sigma^2$$

- We can’t compute either $p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right)$ or $p\left(y_* | \mathbf{y}\right)$ analytically

One alternative approach:

1. Perform analytically computable integrations (over **w**), then
2. Approximate remaining terms and integrals using Type-II maximum likelihood

(Other approximations are available ... see the next lecture and also next semester.)

Slide 7 **Type-II Maximum Likelihood (1)**

- The desired posterior, $p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right)$, can be written as (product rule):

$$p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right) \equiv p\left(\mathbf{w} | \mathbf{y}, \alpha, \sigma^2\right) p\left(\alpha, \sigma^2 | \mathbf{y}\right)$$

- First term is the weight posterior: $p\left(\mathbf{w} | \mathbf{y}, \alpha, \sigma^2\right) = N\left(\mathbf{w} | \boldsymbol{\mu}, \boldsymbol{\Sigma}\right)$

- Second term $p\left(\alpha, \sigma^2 | \mathbf{y}\right)$ is intractable:
  - we approximated it (very coarsely!) with a δ-function at its mode
- We found most probable values $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ which maximised:

$$p\left(\alpha, \sigma^2 | \mathbf{y}\right) = \frac{p\left(\mathbf{y} | \alpha, \sigma^2\right)\, p\left(\alpha\right)\, p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

Slide 8 Type-II Maximum Likelihood (2)

- The posterior to be maximised was:

$$p\left(\alpha, \sigma^2 | \mathbf{y}\right) = \frac{p\left(\mathbf{y} | \alpha, \sigma^2\right)\, p\left(\alpha\right)\, p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

- We assumed flat uninformative priors over $\log \alpha$ and $\log \sigma$:
  - they are scale parameters: this is a very reasonable assumption!
  - and so, we only needed to maximise $p\left(\mathbf{y} | \alpha, \sigma^2\right)$
- $p\left(\mathbf{y} | \alpha, \sigma^2\right)$ is the marginal likelihood (or "evidence")
- This procedure is known as Type-II maximum likelihood

Slide 9 Type-II Maximum Likelihood (3)

- To find $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ we maximised the marginal likelihood, given by:

$$\begin{aligned} p\left(\mathbf{y} | \alpha, \sigma^2\right) &= \int p\left(\mathbf{y} | \mathbf{w}, \sigma^2\right)\, p\left(\mathbf{w} | \alpha\right)\, d\mathbf{w} \\ &= \left(2\pi\right)^{-\frac{N}{2}} \left|\sigma^2 \mathbf{I} + \alpha^{-1} \mathbf{\Phi} \mathbf{\Phi}^{\mathrm{T}}\right|^{-\frac{1}{2}} \exp\left\{ -\frac{1}{2} \mathbf{y}^{\mathrm{T}} \left(\sigma^2 \mathbf{I} + \alpha^{-1} \mathbf{\Phi} \mathbf{\Phi}^{\mathrm{T}}\right)^{-1} \mathbf{y} \right\} \end{aligned}$$

- This is a zero-mean Gaussian distribution over the single N-dimensional dataset vector $\mathbf{y}$ with $N \times N$ covariance matrix:

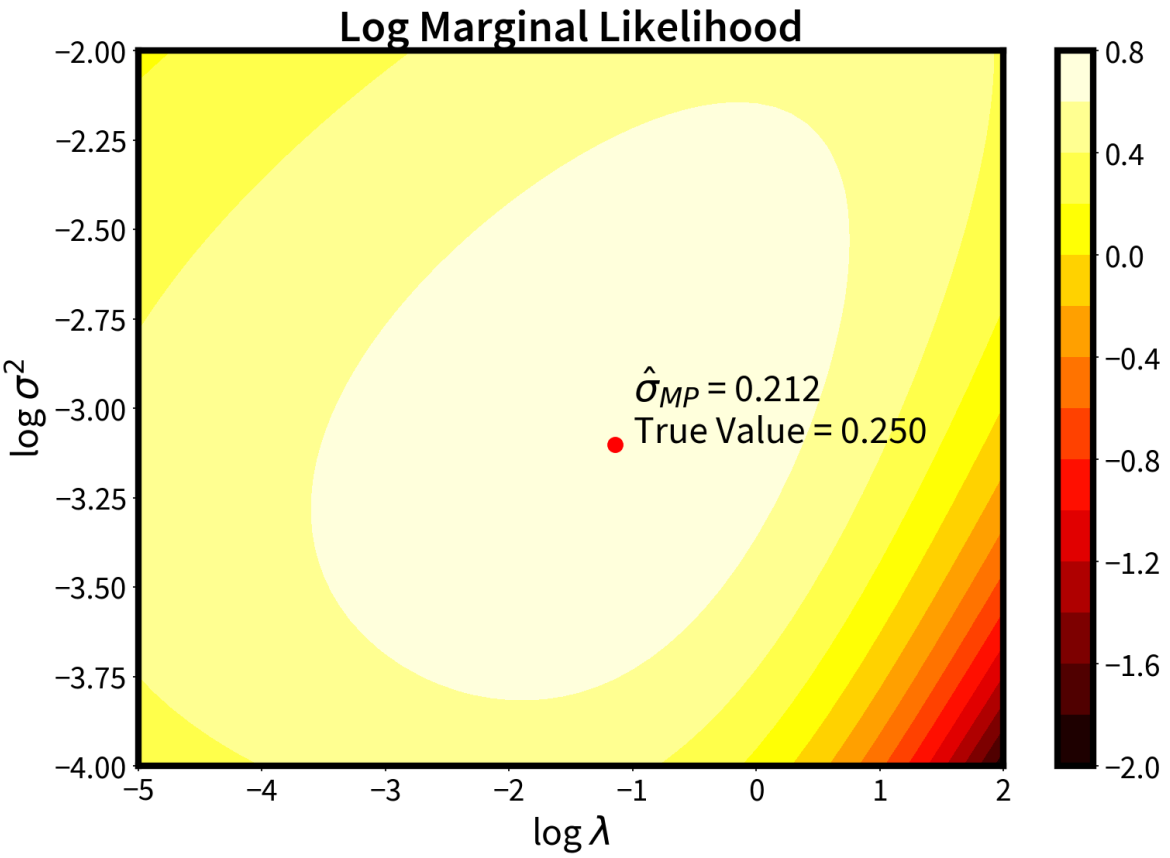
$$\sigma^2\mathbf{I} + \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^{\mathrm{T}}$$

- It gave us a probability for the entire data set, conditional on values of α and σ^2

Slide 10 In Summary: What We Did

- We exploited the "Occam's Razor" property of the marginal likelihood $p\left(\mathbf{y}|\alpha,\sigma^2\right)$ as a criterion first for "setting" α, then later for setting σ
- Advantage: this seemed to work rather well!
- Disadvantage: this only gave us point estimates $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ for the hyper-parameters
- Also, other uncertainties were ignored

Slide 11 Concurrent Estimation (Zoomed In)



Extending the Analysis

Slide 12 A Possible Next Step

- Our estimates of $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ seemed effective
- But, other uncertainties were ignored:
  - the choice of model (linear in the parameters)
  - Gaussian data-centred basis functions ϕ:

$$\phi_m(x) = \exp\left\{ -(x - x_m)^2 / r^2 \right\}$$

- radius (length scale) of the basis functions was fixed at $r = 1.0$

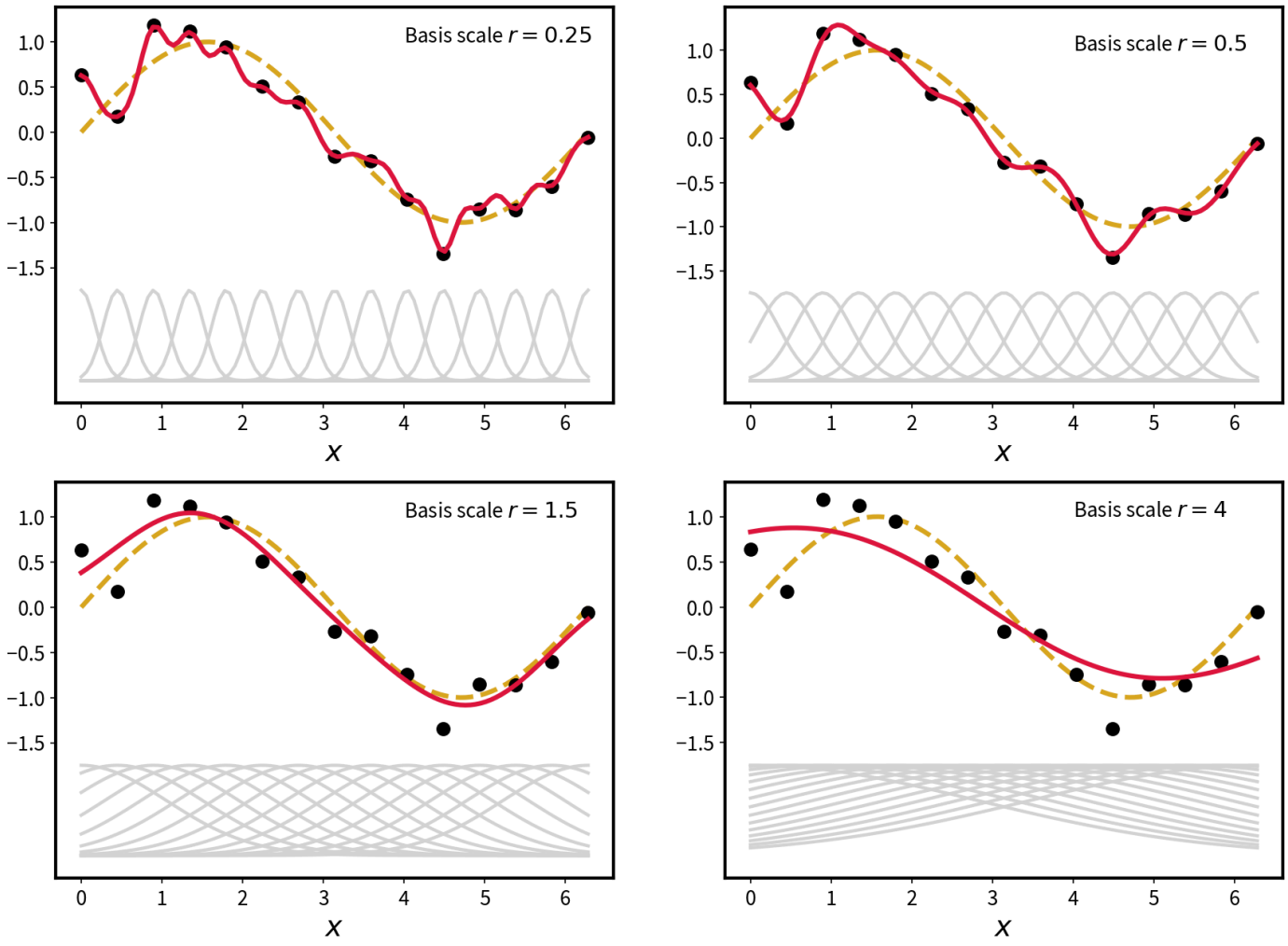
- We can be more Bayesian! Here, it is feasible to *numerically* integrate out α and σ^2 using straightforward sampling techniques (gridding the space)

- We now utilise this approach to find a most probable value for basis scale r

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Slide 13 **Effect of the Basis Function Scale Parameter r**

Illustration of effect of basis scale r (α set to 0.1, noise $= 0.4$):



Slide 14 Type-II Again: Marginalising Further to Take Account of r

We need to integrate out everything but r:

$$\begin{aligned} p\left(\mathbf{y}|r\right) &= \iiint p\left(\mathbf{y}|\mathbf{w},\alpha,\sigma^2,r\right)\, p\left(\mathbf{w}|\alpha\right)\, p\left(\alpha\right)\, p\left(\sigma\right)\, d\mathbf{w}\, d\alpha\, d\sigma \\ &= \iint p\left(\mathbf{y}|\alpha,\sigma^2,r\right)\, p\left(\alpha\right)\, p\left(\sigma\right)\, d\alpha\, d\sigma \end{aligned}$$

Where:

- $p\left(\mathbf{y}|\alpha,\sigma^2,r\right)$ is the marginal likelihood we previously calculated by analytically integrating out $\mathbf{w}$
- $p\left(\alpha\right)$ is a non-informative scale-parameter hyper-prior: constant over $\log\alpha$ for $\alpha \in \left[10^{-12}, 10^{+12}\right]$
- $p\left(\sigma\right)$ is a non-informative scale-parameter hyper-prior: constant over $\log\sigma$ for $\sigma \in \left[10^{-6}, 10^{+2}\right]$

Slide 15 Approximating That Integral!

- We now approximate:

$$p\left(\mathbf{y}|r\right) = \int\int p\left(\mathbf{y}|\alpha,\sigma^2,r\right)\, p\left(\alpha\right)\, p\left(\sigma\right)\, d\alpha\, d\sigma$$

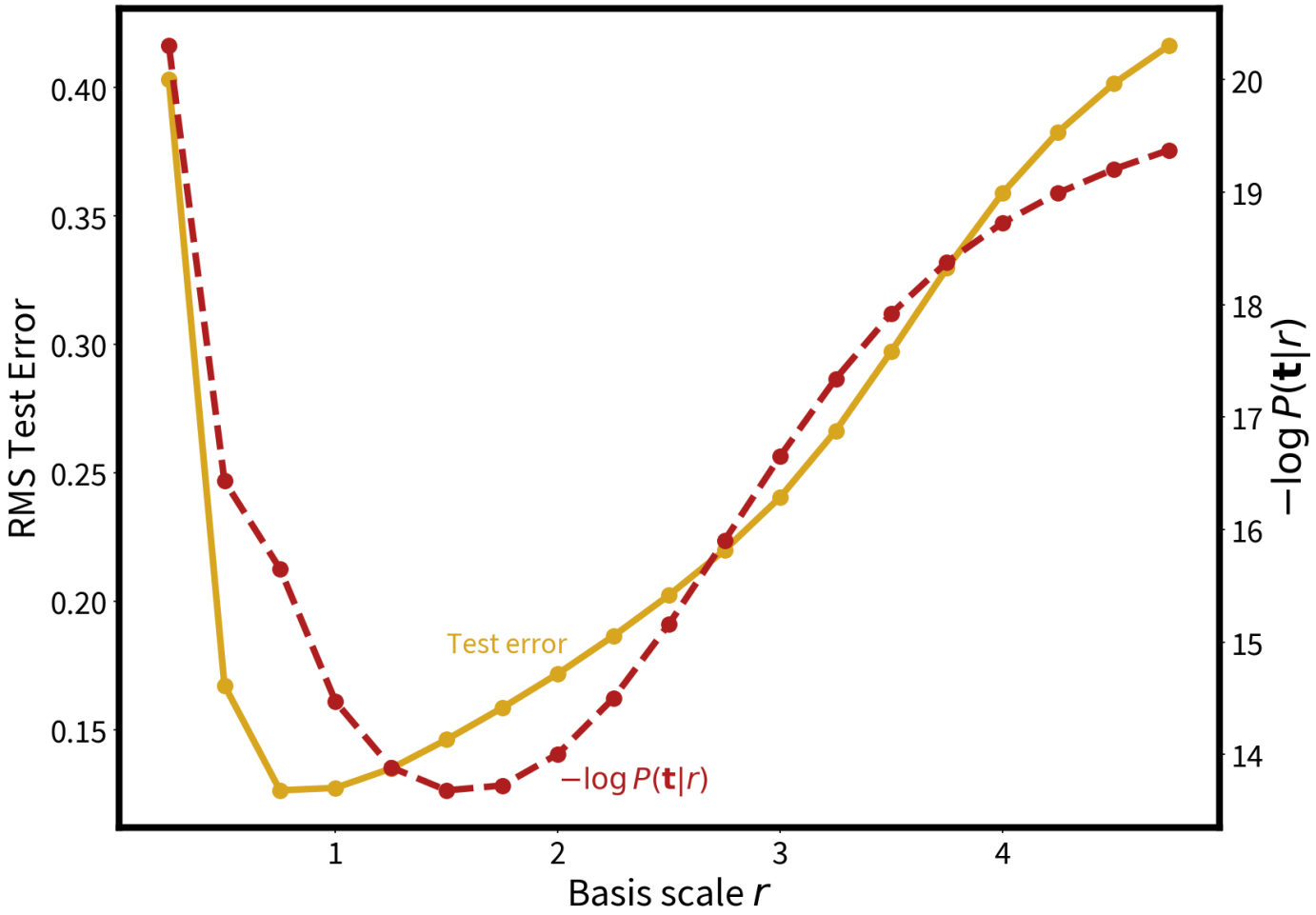
- The integrations over α and σ^2 are performed numerically (and approximately) by a simple Monte-Carlo average (more on these next lecture) *e.g.*:

$$\int p\left(\mathbf{y}|\alpha,\sigma^2,r\right)\, p\left(\alpha\right)\, p\left(\sigma\right)\, d\alpha\, d\sigma \approx \frac{1}{S}\sum_{s=1}^{S} p\left(\mathbf{y}|\alpha_s,\sigma_s^2,r\right),$$

  where α_s and σ_s^2 are representative samples from $p\left(\alpha,\sigma^2\right) = p\left(\alpha\right)\, p\left(\sigma^2\right)$

- In this case, we can simply sample over a regular (uniform) grid of $\log\alpha$ and $\log\sigma$

Slide 16 Computing the Approximation Over a Range of r

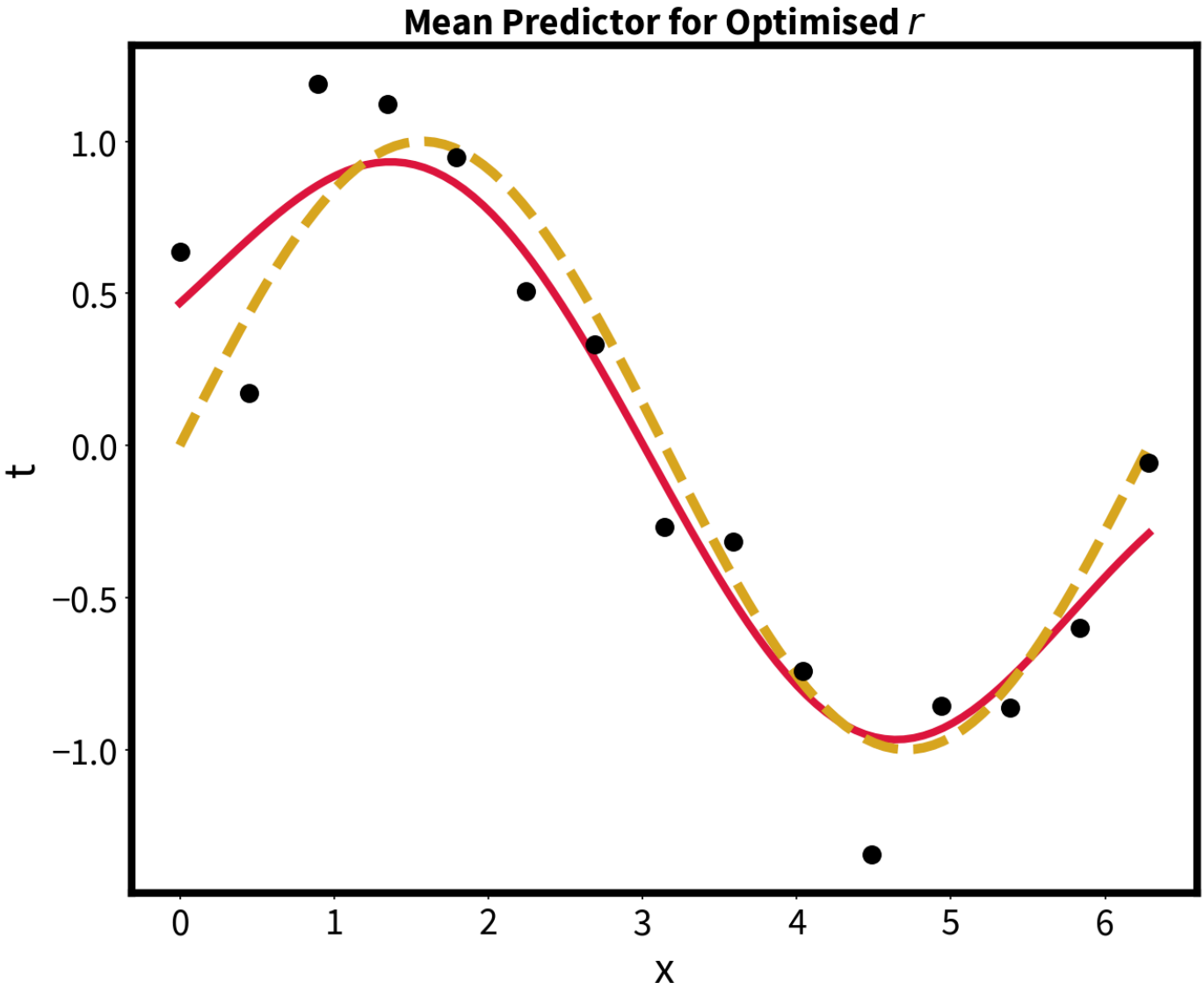


Slide 17 How Can We Predict Using this Approximation?

$$
\begin{aligned}
p\left(y_*|\mathbf{y}\right) &= \iiiint p\left(y_*|\mathbf{w},\sigma^2,r\right)\, p\left(\mathbf{w},\alpha,\sigma^2,r|\mathbf{y}\right)\, d\mathbf{w}\, d\alpha\, d\sigma^2\, dr \\
&\approx \iiint p\left(y_*|\mathbf{w},\sigma^2,\hat{r}_{MP}\right)\, p\left(\mathbf{w},\alpha,\sigma^2|\mathbf{y},\hat{r}_{MP}\right)\, d\mathbf{w}\, d\alpha\, d\sigma^2 \\
&= \iint p\left(y_*|\alpha,\sigma^2,\hat{r}_{MP}\right)\, p\left(\alpha,\sigma^2|\mathbf{y},\hat{r}_{MP}\right)\, d\alpha\, d\sigma^2 \\
&\approx \frac{1}{S}\sum_{s=1}^{S} p\left(y_*|\alpha_s,\sigma_s^2,\hat{r}_{MP}\right)\, p\left(\alpha_s,\sigma_s^2|\mathbf{y},\hat{r}_{MP}\right) \\
&\propto \frac{1}{S}\sum_{s=1}^{S} p\left(y_*|\alpha_s,\sigma_s^2,\hat{r}_{MP}\right)\, p\left(\mathbf{y}|\alpha_s,\sigma_s^2,\hat{r}_{MP}\right)\, p\left(\alpha_s\right)\, p\left(\sigma_s^2\right) \\
&= \frac{1}{S}\sum_{s=1}^{S} p\left(y_*|\alpha_s,\sigma_s^2,\hat{r}_{MP}\right)\, p\left(\mathbf{y}|\alpha_s,\sigma_s^2,\hat{r}_{MP}\right)
\end{aligned}
$$

The mean of $p\left(y_*|\mathbf{y}\right)$ is thus a weighted sum of individual predictive means

Slide 18 Predicting with Optimised r



Demonstration and Wrap-Up

Slide 19 Demo: Occam's Razor (and Again)

Three aspects:

1. Estimation of r
    - repeating the above

2. Being a little more strictly Bayesian by adding r to the numerical integration loop, giving the predictive integral:

$$\begin{aligned} p\left(y_*|\mathbf{y}\right) &= \iiint p\left(y_*|\mathbf{w},\sigma^2,r\right)\, p\left(\mathbf{w},\alpha,\sigma^2,r|\mathbf{y}\right)\, d\mathbf{w}\, d\alpha\, d\sigma^2\, dr \\ &\approx \frac{1}{S}\sum_{s=1}^{S} p\left(y_*|\alpha_s,\sigma_s^2,r_s\right)\, p\left(\alpha_s,\sigma_s^2,r_s|\mathbf{y}\right) \end{aligned}$$

3. Choosing the type of basis function?
    - approximating $p\left(\mathbf{y}|\phi\right)$

Slide 20 Summary

- For three hyper-parameters, it was just about reasonable to compute analytically intractable integrals numerically (summing over a grid)
- This worked pretty well!
- It was possible to obtain accurate estimates of the underlying function and handle the uncertainties in α, σ and r without need for cross-validation
- But this won't work when models become more complex (more parameters and non-linearity)
- Obvious reason: gridding higher-dimensional parameter spaces will become too computationally complex
- Less-obvious reason: the curse of dimensionality